Master Build Prompt: Quotation ERP (Khandelwal Plastic Industries + Satyam Polymers)

Oct 27, 2026 · @Kpi

1. Instructions for the AI: your role and how you must work

Note for the owner (the AI can ignore this note): paste this whole document as your first message into your AI coding tool, then write 'Start with Phase 1'. After each phase, run the checks listed for it and reply 'continue'. If you want no stops, write 'Run all phases without stopping'; the AI must still run every check and give every report.

1.1 Your role

You are a senior full-stack engineer, UI/UX designer and DevOps engineer. Your job is to take this project from an empty folder to a live, tested, deployed production application that I, a non-technical business owner, can operate every day. Everything in this document is required unless it is marked 'optional'.

1.2 Working rules (follow all of them)

1. Work through Phases 0 to 10 (sections 11 and 12) in order. Never skip, merge or reorder phases. Finish and verify one phase before starting the next.
2. At the end of every phase, stop and post a **Phase Report** with: (a) what was built, (b) the files created or changed, (c) how to run it, (d) the real results of automated tests and checks, (e) a checklist for me to test by hand, (f) known limitations, (g) the line 'Reply continue to start Phase N'. Then wait.
3. **Zero placeholders in code.** No TODO, no 'rest of the code here', no '...', no stub functions, no commented-out pseudo-code. Every file must be complete and runnable.
4. **File delivery.** If your tool can create files, create them in the project and make one git commit per phase with a clear message (for example `phase-3: calculation engine and tests`). If you can only reply with text, give one file per code block with its full path as a heading. When your output limit is near, stop at a clean file boundary and write `CONTINUE FROM: path/of/next/file`, then wait for me.

5. **Do not change any rule in this document without asking me.** If something is unclear, ask all your questions in one numbered list at the start of the phase (maximum 8), each with your recommended answer. If I do not answer, use your recommendation and record it in the Phase Report.
6. **Never invent business facts**: company details, GSTIN, PAN, bank details, Terms and Conditions wording, HSN codes or tax rates. Put clearly marked placeholders in seed data (text that starts with `[PLACEHOLDER]` or `[SAMPLE]`) which I replace in Settings. The app must warn me when such text is about to be printed (rule 6.13).
7. **Actions you cannot do yourself** (creating accounts, clicking inside dashboards, buying a domain): give click-by-click steps in plain English with exact button names, then wait for my confirmation. **Actions you can do with the tools you have**: do them and tell me what you did.
8. **Secrets.** Never ask me to paste passwords or secret keys into the chat. Tell me which settings page to enter them in. Never commit secrets. Provide `.env.example` and add an automated secret-scan check.
9. **Language to me.** Write instructions to me in simple English: short sentences, numbered steps, exact button names, and a one-line explanation for any technical word.
10. **Quality bar.** TypeScript in strict mode, ESLint and Prettier clean, every server input validated with Zod, WCAG AA colour contrast, full keyboard use, and a loading state, an empty state and an error state on every screen.
11. **Verification, not claims.** Say a check passed only if you ran it and saw it pass. If you could not run something (for example a real printer test), say so and list it under 'I must test'.
12. **Formats everywhere.** Currency ₹, dates DD/MM/YYYY, time zone Asia/Kolkata, Indian digit grouping (1,23,456.00), financial year April to March, English interface.

2. What we are building and the rules that never change

Build a **Quotation ERP** web application for a plastic packaging business in India (LDPE, HDPE, LLDPE, PP, BOPP, CPP, PET and Nylon bags, films, rolls and tapes). I am the only user. I log in once and work on **two separate companies** inside the same app.

2.1 The two companies

	Company A	Company B
Legal and display name	Khandelwal Plastic Industries	Satyam Polymers
Default quote prefix (editable in Settings)	KPI	SP
Theme name	Ocean Aurora (blue, teal, green)	Rose Silk (light red, cream)
PDF design	Design A (section 9.2)	Design B (section 9.3)

2.2 The isolation rule (most important rule)

A customer who receives a quotation from one company must never see any trace of the other company. A trace means: name, logo, GSTIN, PAN, address, phone, email, bank details, seal, signature, colours, illustrations, fonts, quote prefix, file name, PDF title and author metadata, watermark text, and the names of embedded image files. Build an automated test that extracts the text and metadata of every generated PDF and fails if any string of the other company appears (see section 13.9).

2.2.1 What is shared and what is separate

Data	Shared or separate
Login	One login for both companies
Company profile, bank, logo, seal, signature, PDF theme	Separate per company
Quote number series	Separate per company and per financial year
GST type (CGST+SGST or IGST)	Decided per quotation from that company's state
Customer master	Shared (one customer can buy from both)
Item master	Shared; **the rate is separate per company and per unit**

Data	Shared or separate
T&C presets	Shared list
A customer's default T&C	Separate per customer and per company
Dropdown options	Shared
Dashboard, reports, Recycle Bin, backup export	Filterable by company: KPI, SP or All

2.3 Scope of version 1

- Included: login, Home, Dashboard, quotation create, edit, revise, duplicate, delete and restore, status tracking, conversion to PO, two A4 PDFs, customer master, item and rate master, T&C presets, reports, settings, backup export, deployment.
- Not included: AI features, invoices, e-invoicing, e-way bills, multiple users, roles, payments. Do not build them. But store HSN code and place of supply now so invoices can be added later without changing the database.
- One owner account only. No sign-up page. No roles.

2.4 Non-negotiable rules

1. Money is stored and calculated as integers (paise). Never use floating point for money.
2. A saved quotation keeps its own snapshot of customer details, item details, rates and T&C. Changing a master record never changes an old quotation.
3. Quotation numbers are generated atomically in the database and are never reused.
4. Dashboards count only the latest revision of each quotation.
5. A Converted quotation is locked. Deleting is always a soft delete into the Recycle Bin.
6. Every table has Row Level Security. The service-role key never reaches the browser.
7. The company of a quotation is locked once it is saved.
8. Rate unit is always the same as quantity unit (section 6.2).

3. Technology stack and project structure

Use the stack below. Use the latest stable version of each package at build time, pin exact versions in `package.json`, and run `npm audit` before every release. Do not swap a technology without asking me.

3.1 Stack

Layer	Choice	Notes
Framework	Next.js (App Router), React, TypeScript strict	Server actions or route handlers for all data writes
Styling	Tailwind CSS, CSS variables for themes	Theme switches through a `data-company` attribute on the root element
Icons	lucide-react	
Charts	Recharts	Gradient fills, animated draw-in
Animation	Framer Motion plus CSS transitions	Respect `prefers-reduced-motion`
Forms and validation	React Hook Form and Zod	One Zod schema shared by client and server
Database, auth, files	Supabase: Postgres, Auth, Storage, region Mumbai (ap-south-1)	Row Level Security on every table
Hosting	Vercel, function region near Mumbai (bom1)	
PDF	HTML and CSS A4 templates (section 9)	See 3.2
Excel import and export	ExcelJS or SheetJS	
Unit and integration tests	Vitest and React Testing Library	
End-to-end tests	Playwright	
Lint, format	ESLint, Prettier, TypeScript strict	CI blocks merges on failure
Error tracking	Sentry (optional, free tier)	

3.2 PDF method (decide in Phase 5, with this rule)

Build each quotation as an HTML and CSS React component sized exactly 210 mm by 297 mm. Produce the PDF in two ways from the same component: (1) a server-side headless Chromium render (for example `puppeteer-core` with `@sparticuz/chromium` in a Vercel function) with background graphics on; (2) a browser Print fallback using `@page { size: A4; margin: 0 }` and `print-color-adjust: exact`. Test (1) on Vercel limits. If it works reliably, it is the primary method; if not, make (2) primary and tell me why. Do not use html2canvas, because it produces blurry, non-selectable PDFs.

3.3 Folder structure (create this and keep it)

```
quotation-erp/
  app/
    (auth)/login/
    (app)/home/
    (app)/dashboard/
    (app)/quotations/           list, [id] view, new, [id]/edit
    (app)/customers/
    (app)/items/
    (app)/tc-presets/
    (app)/reports/
    (app)/recycle-bin/
    (app)/settings/
    api/pdf/[id]/route.ts
    api/cron/expire-quotes/route.ts
  components/
    shell/        sidebar, topbar, company-switcher, command-palette
    ui/           buttons, inputs, searchable-select, toast, skeleton,
dialogs
    charts/
    quotation/    form, item-row, totals-panel, tc-editor, live-preview
    pdf/          design-a (Khandelwal), design-b (Satyam), shared helpers
    illustrations/
  lib/
    domain/       money, indian-format, amount-in-words, gst, size, units,
fy, numbering, validators, status
    supabase/     browser client, server client, admin client (server only)
    auth/
    export/
  supabase/
    migrations/
```

```
    seed/
  tests/
    unit/  integration/  e2e/  fixtures/
  docs/           README.md, ARCHITECTURE.md, RUNBOOK.md, CHANGELOG.md
  .env.example
```

3.4 Engineering conventions

- All business logic lives in `lib/domain` as pure functions with no framework or database imports, so it can be tested alone.
- The quotation total is computed by the same domain function in the browser (live preview), on the server (before saving) and in tests. The server value is the one stored.
- Dates are stored as `date` or `timestamptz` in UTC and displayed in Asia/Kolkata as DD/MM/YYYY.
- Every list is paginated on the server. Search boxes are debounced by 250 ms.
- Every write goes through a server function that validates input with Zod and writes an audit log entry.

4. Database model and data security

Create all tables through versioned SQL migrations in `supabase/migrations`. Every table has `id` (uuid), `created_at`, `updated_at`; business tables also have `deleted_at` (soft delete). Add sensible indexes, foreign keys and CHECK constraints. The columns below are the minimum; add more if needed and tell me.

4.1 Tables

Table	Purpose	Key columns
`app_users`	The single owner	auth_user_id, login_id (unique), display_name, recovery_email

Table	Purpose	Key columns
`companies`	KPI and SP profiles	code, legal_name, display_name, address lines, city, state, state_code (2 digit GST code), pin, phones, email, website, gstin, pan, quote_prefix, logo_path, seal_path, signature_path, pdf_theme, default_validity_days, round_off_enabled, is_active
`company_bank_accounts`	Bank details per company	company_id, bank_name, account_name, account_number, ifsc, branch, is_default
`customers`	Shared customer master	name, contact_person, phone, email, address, city, state, state_code, pin, gstin, notes, is_active
`items`	Shared item master	code (unique, case-insensitive), name, specification, material, default sizes and units, spec_value, spec_unit, print_type, print_colours, hsn, default_gst_rate, default_unit, is_active
`item_rates`	Rate per item, per company, per unit, with history	item_id, company_id, unit, rate_e4 (integer, see 6.1), is_current, effective_from, changed_by_source (manual or quote), note

Table	Purpose	Key columns
`quotations`	One row per revision	company_id, series_no, fy_label, quote_number, revision_no, base_quote_id, is_latest_revision, status, quote_date, validity_days, valid_until, customer_id, customer snapshot columns (name, contact, phone, email, address, city, state, state_code, pin, gstin), delivery_address, requirement_via, reference, subject, place_of_supply_state_code, tax_mode (INTRA or INTER), subtotal_paise, discount_type, discount_value, discount_paise, freight_label, freight_paise, freight_gst_rate, taxable_paise, cgst_paise, sgst_paise, igst_paise, round_off_paise, net_paise, amount_in_words, remarks, internal_note, sent_at, converted_at, row_version
`quotation_items`	Item lines with snapshot	quotation_id, line_no, item_id (nullable), item_code, item_name, specification, material, size1/2/3 value and unit, size_display, spec_value, spec_unit, print_type, print_colours, hsn, qty_milli (integer), unit, rate_e4, gst_rate, line_amount_paise
`quotation_terms`	T&C rows of a quotation	quotation_id, sort_order, heading, body, is_enabled
`quotation_status_history`	Every status change	quotation_id, from_status, to_status, changed_at, note
`tc_presets`	Reusable T&C wording	heading_key, title, body, sort_order, is_active

Table	Purpose	Key columns
`customer_tc_defaults`	A customer's default T&C per company	customer_id, company_id, sort_order, heading, body, is_enabled
`dropdown_options`	Dropdown Manager data	group_key, label, value, sort_order, is_active, is_system
`po_records`	Conversion details	quotation_id, po_number, po_date, po_value_paise, attachment_path, notes
`number_sequences`	Atomic numbering	company_id, fy_label, last_number (primary key company_id plus fy_label)
`audit_log`	Who changed what	company_id, entity_type, entity_id, action, before (jsonb), after (jsonb), created_at
`login_attempts`	Lockout tracking	login_id, ip_address, succeeded, created_at
`app_settings`	Owner settings	inactivity_minutes, followup_days, expiring_soon_days, and similar keys
`states`	Indian states and union territories	name, gst_code, is_ut_without_legislature

4.2 Database functions

- `next_quote_number(company_id, quote_date)`: in one transaction, work out the financial year label from the date, do `INSERT ... ON CONFLICT DO UPDATE SET last_number = last_number + 1 RETURNING`, and return the series number. It must be safe under 50 parallel calls (test it).
- `mark_expired_quotes()`: sets Sent and Negotiation quotations with `valid_until` before today (Asia/Kolkata) to Expired and writes status history. Called by the daily job (6.8).
- An `is_owner()` helper used by every policy.

4.3 Security rules in the database

1. Enable Row Level Security on every table. Policies allow access only when `is_owner()` is true. Public sign-up is disabled in Supabase Auth.
2. Files live in a **private** Storage bucket (`company-assets`) with an `attachments` folder for PO files. Allow PNG, JPEG and WebP only, maximum 2 MB for brand images and 5 MB for PO attachments (PDF or image). Check the real file type, not just the extension. Show files through short-lived signed URLs (maximum 60 minutes). For PDFs, the server embeds images as data URIs so the PDF is self-contained.
3. The service-role key is used only in server code for: login attempt logging, the lockout check, the cron job, and storage reads for PDFs. It is never sent to the browser and never has a `NEXT_PUBLIC_` name.
4. Prove the policies with an automated test: with the anonymous key and with no session, every table returns zero rows and every write fails.

4.4 Seed data (Phase 2)

- Both companies with clearly marked placeholder details (`[PLACEHOLDER]`), the correct default prefixes, and the correct theme names.
- All Indian states and union territories with GST codes (verify against the official list and tell me the source). Mark the union territories without legislature, because they use UTGST instead of SGST.
- Default dropdown options (6.2), 8 T&C headings (6.11) and sample presets marked `[SAMPLE]`.
- No fake customers, items or quotations in the production seed. A separate `seed:demo` script may add demo data for the development database only.

5. Login and authentication

One owner logs in with a **Login ID and a passcode** and lands on the Home page. There is no sign-up and no role system.

1. **Accounts.** Use Supabase Auth with email and password. Store the owner's real email as the Auth email. The `app_users` table maps the Login ID to that user. The login form takes Login ID and passcode; the server looks up the email and signs in. The email is never shown on the login page.

2. **Passcode rules.** Minimum 10 characters. Settings has 'Change passcode' (asks for the current one). 'Forgot passcode' sends a Supabase reset email to the owner's recovery email. Document the manual recovery route in `RUNBOOK.md`.
3. **Lockout.** After 5 wrong attempts within 15 minutes for a Login ID (or from one IP), lock sign-in for 15 minutes. Enforce this on the server using `login_attempts`, not in the browser. Show a countdown. Always show the same message for a wrong Login ID or a wrong passcode ('Login ID or passcode is incorrect') so nobody can discover valid IDs.
4. **Sessions.** Use secure, httpOnly cookies through `@supabase/ssr`. Auto-logout after 30 minutes of no activity (setting 5 to 120 minutes), with a warning dialog one minute before. Maximum session length 7 days. Logout clears the session on all tabs.
5. **Route protection.** Middleware redirects every route except the login page to login when there is no valid session. After login, go to Home. Remember the page the owner was heading to and return there.
6. **Login page UX.** Left: brand panel with mesh gradient and an original abstract plastic-film and roll illustration (blend of both company palettes). Right: glass card with Login ID, passcode with show or hide eye, 'Remember my Login ID' (stores only the ID, never the passcode), 'Forgot passcode' link, a loading state on the button and a clear lockout message. Card entrance animation: fade and slide up, 300 ms.
7. **Security headers and input handling.** Set Content-Security-Policy, Strict-Transport-Security, X-Frame-Options DENY, X-Content-Type-Options and Referrer-Policy. Rate limit the login action. All text fields (customer name, T&C, remarks) are plain text: escape on render, never use `dangerouslySetInnerHTML`, and strip control characters on save.
8. **Audit.** Log successful logins, failed logins, lockouts, passcode changes and logouts in `audit_log`.

6. Business logic (implement exactly as written)

This section is the source of truth for every calculation and rule. Put the calculation parts in `lib/domain` with tests (section 13).

6.1 Money, rates and quantities

- Amounts are integers in paise. Quantity is stored as an integer in thousandths (`qty_milli`: 5000 Kg is 5,000,000). Rate is stored with up to 4 decimals as `rate_e4` (₹142 is 1,420,000; ₹0.375 is 3,750), because small items are often priced per piece in fractions of a rupee.

- Line amount in paise = `qty_milli × rate_e4 / 100000`, rounded half up to the nearest paisa. Use BigInt in code so large values never overflow.
- The only rounding mode anywhere is round half up, implemented once in one helper.
- Display amounts with 2 decimals in Indian grouping. Display rates with at least 2 and at most 4 decimals. Display quantities with up to 3 decimals, trailing zeros trimmed. Input fields accept commas and the ₹ sign and clean them.

6.2 Units and default dropdown lists

Every fixed-choice field is a searchable dropdown with a '+ Add new option' action (a small dialog that saves to `dropdown_options`). Options that are in use can be disabled but never hard-deleted. A quotation stores the label text, so old quotations always print correctly. Seed these groups:

Group	Default options
Requirement Via	Mail, WhatsApp, Person, Phone Call, IndiaMART, Reference, Other
Quantity unit	Kg, Pcs, Nos, Box, Rolls, Meter, Sq. Meter
Size unit	mm, cm, inch, m
Spec unit	GSM, Micron, Gauge
Material	LDPE, HDPE, LLDPE, PP, BOPP, CPP, PET, Nylon
Print	Plain, Printed (colours 1 to 8)
Validity	7 Days, 15 Days, 30 Days, Custom Date
GST rate	0, 5, 18 (all editable; never hard-code tax slabs; the owner and CA decide)
Lost reason	Price too high, Competitor chosen, Requirement cancelled, No response, Other
T&C heading	The 8 headings in 6.11

Rate unit rule: the rate unit is always the same as the quantity unit. If quantity is in Kg, the rate is ₹ per Kg; Sq. Meter gives ₹ per Sq. Meter; Pcs, Nos and Box give ₹ per Pcs, per Nos and per Box. There is no separate rate-unit field. The rate label updates automatically when the quantity unit changes.

6.3 Merged size

The form has Size 1 and Size 2 (each value plus unit) and an optional Size 3. A read-only box shows the merged text and updates live. Rules:

- Only Size 1 filled: `44 mm`. Size 1 and Size 2 filled: `44 mm x 44 mm`. With Size 3: `350 mm x 450 mm x 50 mm`. Each part keeps its own unit.
- Use a lowercase 'x' with single spaces. Remove trailing zeros (44.50 becomes 44.5).
- Size 2 filled while Size 1 is empty is a validation error. Every filled size needs a value above 0 and a unit.
- Store the merged text in `size_display`.

6.4 GST and totals engine

Calculate in this exact order:

1. Line amounts (6.1). **Subtotal** is their sum.
2. **Discount**: percentage (0 to 100, 2 decimals) or a fixed ₹ amount; never more than the subtotal. A percentage is rounded half up to paise.
3. **Allocate the discount** across GST-rate groups in proportion to each group's line total, using the largest-remainder method so the parts add up exactly to the discount. Ties go to the lower GST rate.
4. **Freight and packing**: optional ₹ amount with an editable label and its own GST rate (default: the highest rate used on the lines). It forms its own group and is not discounted. In the Phase 3 report, tell me that my CA must confirm this treatment.
5. **Taxable total** = subtotal − discount + freight.
6. **GST per rate group.** Intra-state: CGST = round half up of (group taxable × rate ÷ 2 ÷ 100), SGST is the same amount. Inter-state: IGST = round half up of (group taxable × rate ÷ 100). Show every rate on its own line.
7. **Gross** = taxable total + all GST.
8. **Round-off** (if the company setting is on): round gross to the nearest rupee (half up). Round-off = rounded − gross; it can be negative and is shown as (+) or (−).
9. **Net Amount** = gross + round-off.
10. **Amount in words** from the Net Amount (rules below).

Intra or inter-state. Compare the company's state code with the quotation's **Place of Supply** state code. Place of Supply defaults to the first two digits of the customer's GSTIN, otherwise the customer's state, and the owner can change it. Equal means INTRA

(CGST + SGST); different means INTER (IGST). With no Place of Supply the quotation can be saved as a Draft but not marked Sent. For union territories without legislature, label the state-level tax UTGST instead of SGST (driven by the `states` table flag).

Amount in words (Indian system). Use Thousand, Lakh and Crore. Examples: `Rupees Forty Four Thousand Only`; with paise `Rupees One and Fifty Paise Only`; paise only `Fifty Paise Only`; zero `Rupees Zero Only`. Title Case. Support at least up to 999 crore.

6.5 Financial year and quotation numbers

- Financial year runs 1 April to 31 March and is labelled `2026-27`. It is taken from the quote date.
- Quote number = `PREFIX/FY/NNNN`, for example `KPI/2026-27/0001`. The sequence number is zero-padded to 4 digits and grows naturally above 9999. A revision adds `-R1`, `-R2` (`KPI/2026-27/0001-R1`). Revision 0 has no suffix in the number but shows 'R0' in the Rev No field.
- Each company has its own series per financial year, restarting at 0001 on 1 April.
- The number is assigned at the first server save (6.12), generated by the database function, and never reused, even if the draft is later deleted. Gaps can therefore occur; that is acceptable for quotations. Before the first save the form shows 'Will be assigned when you save'.
- Changing the quote date into another financial year after saving does not change the number; show a warning.
- Changing a company's prefix in Settings affects only future quotations; warn me.
- PDF file name: `KPI-2026-27-0001-R1_CustomerName.pdf` (slashes become hyphens; the customer name keeps only letters, digits and underscores; maximum 100 characters).

6.6 Revisions

- **Revise** is available on the latest revision of a Sent, Negotiation, Lost, Cancelled or Expired quotation. A Draft is edited directly. A Converted quotation is locked. A Superseded one is view-only.
- Revise creates a new row that copies everything (customer snapshot, items, rates, terms, charges), with revision_no + 1, the same series number and base_quote_id, status Draft, quote date today (editable) and validity reset to the company default. The old revision becomes **Superseded** and `is_latest_revision` flips, in one database transaction. Old data stays untouched and viewable.

- Lists show only latest revisions. Each row can expand to show R0, R1, and so on, with a 'Show all revisions' toggle.
- Dashboards, KPIs, reports and search count only rows where `is_latest_revision` is true.
- If a latest revision is deleted, ask whether to reinstate the previous revision (its earlier status is restored from the status history).

6.7 Status lifecycle

Statuses: Draft, Sent, Negotiation, Converted, Lost, Cancelled, Expired, Superseded.

From	Allowed next status	Rules
Draft	Sent, Cancelled	Edit freely. Marking Sent needs full validation (6.13) and sets `sent_at`
Sent	Negotiation, Converted, Lost, Cancelled	Expired is automatic (6.8); Superseded happens only through Revise
Negotiation	Converted, Lost, Cancelled	Same automatic rules as Sent
Converted	none	Locked. Only the special 'Reverse conversion' action (needs a reason and typing the quote number) returns it to Negotiation
Lost	none	Lost reason is required. Use Revise to quote again
Cancelled	none	Optional note. Use Revise to quote again
Expired	none	Automatic only. Use Revise to quote again
Superseded	none	Automatic only

- **Editing rules.** Draft: edit freely. Sent or Negotiation: the Edit button first asks 'This quotation was already sent. Create a revision instead?' with 'Revise (recommended)' and 'Edit anyway' (logged in the audit log). Converted, Lost, Cancelled, Expired and Superseded: view only.
- Reject illegal transitions on the **server**, not only in the interface, and test this.
- Every transition writes a `quotation_status_history` row and an `audit_log` row.

6.8 Validity and automatic expiry

- `valid_until` = quote date + 7, 15 or 30 days, or the custom date. The quotation is valid through that date (inclusive).
- Sent and Negotiation quotations become **Expired** when `valid_until` is before today (Asia/Kolkata). Drafts never expire.
- Run a daily job at 00:15 IST: Supabase `pg_cron` calling `mark_expired_quotes()` (preferred), or a Vercel Cron call to `/api/cron/expire-quotes` protected by `CRON_SECRET`.
- Fallback: lists and dashboards also treat a Sent or Negotiation quotation with a past `valid_until` as Expired, and the app runs `mark_expired_quotes()` after each login, so nothing depends on the job alone.

6.9 Delete, Recycle Bin and locking

- Delete always asks for confirmation and shows the quote number and customer. It sets `deleted_at`; the record disappears from every screen except the Recycle Bin.
- A Converted quotation cannot be deleted (reverse the conversion first).
- If the quotation has more than one revision, the delete dialog offers two choices: 'Delete only this revision' (follows 6.6) or 'Delete the whole quotation (all revisions)'.
- Recycle Bin lists deleted quotations, customers and items, each with Restore and 'Delete permanently' (permanent delete needs typing the word DELETE). Restoring a quotation brings back its previous status. There is no automatic purge unless I enable one in Settings.
- A customer or item that is used by any quotation can be deactivated or soft-deleted but never permanently deleted.

6.10 Converting a quotation to a PO

The 'Convert to PO' action opens a dialog with: PO number (required), PO date (required, DD/MM/YYYY, not earlier than the quote date), PO value in ₹ (defaults to the quotation's Net Amount, editable), an optional attachment (PDF or image, maximum 5 MB) and optional notes. On save: create the `po_records` row, set status Converted, set `converted_at`, lock the quotation, and write status history and audit rows. The PO value, not the quoted value, is the converted value in dashboards (6.14).

6.11 Terms and Conditions

- Every quotation has **8 T&C rows by default** and a maximum of 12 ('Add row').
- Each row has: a Heading dropdown (searchable, with '+ Add new heading'), a Body text area, a **Preset dropdown** filtered by the heading (choosing a preset fills the body, which stays fully editable), an On/Off checkbox, and move up or down buttons. Rows that are Off or empty are not printed. Printed rows are numbered 1, 2, 3 without gaps. Maximum 1000 characters per row.
- The 8 default headings, in order: Payment Terms, Delivery and Lead Time, Quantity Tolerance, Freight and Transport, GST and Taxes, Price Validity, Packing, Jurisdiction.
- A new quotation loads its rows from: (1) this customer's saved default for this company, else (2) the company's standard preset for each heading, else (3) the 8 headings with `[SAMPLE]` text.
- 'Save as this customer's default for this company' stores the current rows in `customer_tc_defaults`. Customers can have different terms with each company.
- The T&C Presets page lets me add, edit, reorder, disable and delete presets for each heading, and mark one preset per heading and company as the standard.
- Do not write legal or commercial wording yourself. Seed only short sample text starting with `[SAMPLE]`; I will replace it.

6.12 Saving, drafts, items, rates and customers

Saving and drafts

- Before the first server save, keep the form in browser local storage for crash recovery.
- The first server save happens when I press Save Draft, or automatically 2 seconds after I have entered a customer name and at least one item with a quantity. At that moment the quote number is generated.
- After the first save, autosave runs 2 seconds after the last change and shows 'Saving...', 'Saved at 10:42' or 'Not saved, retry'. Use `row_version` for optimistic concurrency; if another tab or device saved newer data, warn and offer to reload.
- Warn before leaving the page or navigating away with unsaved changes.

Items and rates

- Item Code and Item Name are searchable pickers over the active item master. Each suggestion shows code, name, material, size and this company's rate. Choosing one fills the item snapshot and the rate for the selected quantity unit.

- A code that is not in the master: on save, ask 'Save these new items to the master?' (a checklist). Saving creates the item and this company's rate for that unit.
- Details of an existing item changed inside the quotation: on save, ask 'Update the item master?' (default No).
- Rate different from the master rate: on save, ask 'Update the master rate for KPI from ₹142 to ₹150 per Kg?' with 'Update master rate' and 'Only this quotation'. Updating inserts a new `item_rates` row and flips `is_current` (history is kept).
- Under the rate field show: 'Master (KPI): ₹142/Kg. Last to this customer: ₹140/Kg on 12/09/2026'. Use only quotations of the same company. If none: 'No earlier quotation to this customer'.
- If the quantity unit on a line changes and the master has a rate for the new unit, fill it; otherwise clear the rate and show 'No master rate for Pcs'.
- Master rates can be edited at any time on the Items and Rates page, with a history table (date, old rate, new rate, source). Changing a master rate never changes an existing quotation.

Customers

- Customer Name autocompletes by name, phone or GSTIN (top 8 results, 250 ms debounce). Choosing a customer fills every customer field as an editable snapshot.
- A new name: on save, ask 'Save as a new customer?'. Changed details for an existing customer: ask 'Update the customer master?' (default No). A GSTIN that already belongs to another customer shows a duplicate warning.

6.13 Validation and print safety

Field	Rule
GSTIN	Converted to upper case; 15 characters matching the standard GSTIN pattern; first two digits must be a valid state code. A failed checksum is a **warning**, not an error
PAN (company)	5 letters, 4 digits, 1 letter
PIN code	6 digits, first digit not 0
Phone	Indian mobile of 10 digits starting 6 to 9 (accept +91, 91 or 0 in front, spaces and hyphens), or a landline with STD code; store normalised digits

Field	Rule
Email	Valid email format
IFSC	4 letters, the digit 0, then 6 letters or digits
Bank account	9 to 18 digits
Numbers	Quantity above 0 (up to 3 decimals); rate above 0 (up to 4 decimals); discount not above subtotal

- **To save a Draft:** company and customer name.
- **To mark Sent (or print a final PDF):** company, quote date, validity, customer name, address, state or Place of Supply, at least one phone or email, at least one item with name, quantity, unit, rate and GST rate, and text in every enabled T&C row. Show all errors inline next to each field and a summary at the top.
- **Print safety.** Block a non-Draft PDF when the company profile still contains `[PLACEHOLDER]` text or has no GSTIN, address or bank details, and show 'Complete the company profile in Settings' with a link. Warn (do not block) when any T&C row still contains `[SAMPLE]`. A Draft PDF is always allowed and carries the DRAFT watermark.

6.14 Dashboard and report definitions

All figures use: latest revisions only, deleted rows excluded, quote date inside the chosen range, and the chosen company (KPI, SP or All).

Metric	Definition
Total quotations	Count and sum of Net Amount of all statuses, Drafts included (show a small 'includes N drafts' note)
Converted to PO	Count; value = sum of PO values (tooltip also shows the quoted value); win rate below
Win rate	Converted ÷ (Converted + Lost + Cancelled) × 100, one decimal. Show a dash when the denominator is 0. The tooltip shows the formula and the three counts
Pending	Sent + Negotiation: count and Net Amount

Metric	Definition
Lost or Cancelled	Count and Net Amount of both together; a small line shows Expired count and value
Expiring soon	Sent or Negotiation with `valid_until` within the next 3 days (setting): count and value
Average quotation value	Total value ÷ total count
Funnel	Created (all counted quotations), Sent (ever reached Sent, from status history), Converted

- Date presets: Today, Last 7 Days, This Month (calendar), This FY (1 April to 31 March), Custom (inclusive).
- Trend chart buckets: daily when the range is 62 days or less, otherwise monthly. Series: created and converted.
- Company comparison chart (grouped bars) appears only when the company filter is All.
- Customer-wise table columns: customer, quotations, total value, converted count, converted value, win rate, last quotation date. Sortable, searchable, paginated, exportable to Excel.
- Home 'Today's work' lists: expiring within 3 days; Sent for 7 or more days with no status change (setting); Drafts untouched for more than 1 day.

7. Application shell, navigation and every page

Home and Dashboard are **two separate pages**. Every list has search, sort, server-side pagination, skeleton loading, an illustrated empty state with a call-to-action, an error state with a retry button, and success or error toasts.

7.1 Shell

- **Collapsible sidebar.** Expanded 260 px with icon and name; collapsed 72 px with icons only. Collapsed items show a tooltip with the name; items with sub-menus show a small flyout. A toggle button at the top and the shortcut `Ctrl + B` switch the state, animated over 250 ms, and the content area resizes with it. The chosen state is remembered in the browser (local storage). On the quotation form the sidebar collapses automatically to make room for the live preview and restores the previous

state when I leave. Below 1024 px wide the sidebar is hidden and opens as a drawer from a hamburger button, with a dark overlay and tap-outside-to-close. The sidebar, topbar and every control are never printed.

- **Menu groups.** Main: Home, Dashboard. Sales: Quotations (All, New), Customers. Masters: Items and Rates, T&C Presets. Insights: Reports. System: Recycle Bin, Settings. Active page: company-accent highlight and a thin indicator line. Quotations shows a badge with the Draft count and a red dot when quotations are expiring soon (visible when collapsed too). Bottom of the sidebar: owner name and avatar, dark mode toggle, Logout.
- **Company switcher** at the top of the sidebar with three states: Khandelwal Plastic Industries, Satyam Polymers, and All Companies. The choice sets one global `activeCompany` that every page reads, and it is remembered across refreshes (cookie). Switching changes the `data-company` attribute, so the accent gradient, sidebar top strip, buttons, charts and illustrations change app-wide with a 300 ms colour transition. All Companies uses a neutral indigo accent and is for viewing only (Home, Dashboard, Reports, Quotations list, Recycle Bin). Pressing '+ New Quotation' in All Companies mode first asks 'Which company?'. The Dashboard filter chips (All, KPI, SP) control this same global value.
- **Topbar** (glass effect with blur and a thin gradient line): hamburger on small screens, page title, global search with `Ctrl + K` (a command palette that finds quotations by number, customer or item code, plus customers, items, pages and actions such as 'New quotation'), the primary '+ New Quotation' button, a notification bell (expiring soon and no-reply items), dark mode toggle, profile menu (Change passcode, Logout).
- Themed 404, error and 'session expired' pages.

7.2 Home page (start here every day)

- Gradient greeting banner for the active company: 'Good evening, [name]', today's date and a floating original illustration.
- Four large quick-action buttons: New Quotation, Customers, Items and Rates, Reports.
- **Today's work** list: quotations expiring within 3 days; Sent quotations with no reply for 7 days or more; Drafts left unfinished. Each item shows a company chip, the customer, the amount and one-click actions (Open, Mark Negotiation, Revise).
- Last 8 quotations in a compact list (quote number, customer, amount, status chip, company chip).
- Three small numbers: created today, created this month, pending (Sent plus Negotiation).

- Global search `Ctrl + K`.

7.3 Dashboard (analysis)

- Header controls: company filter and date preset (Today, Last 7 Days, This Month, This FY, Custom range).
- Six KPI cards with count-up numbers, a gradient icon chip, a sparkline, and a tooltip with the exact definition from 6.14: Total quotations, Converted to PO (with win rate), Pending, Lost or Cancelled, Expiring soon, Average quotation value. Large amounts show compactly (₹1.25 Cr, ₹12.5 L) with the full amount in the tooltip.
- Charts (gradient fills, animated draw-in): monthly or daily trend (created and converted), status donut, top 5 customers by value, Quote to PO funnel, and company comparison (All mode only).
- **Customer-wise table** (columns in 6.14) with search, sort, pagination and Excel export.
- Everything recalculates when a filter changes, with skeleton placeholders while loading.

7.4 Quotations list

- Status tabs: All, Draft, Sent, Negotiation, Converted, Lost, Cancelled, Expired. A toggle 'Show all revisions' includes Superseded rows.
- Search by quote number, customer name, item code, item name or reference. Filters: date range, customer, amount range. Sort by any column. 25 rows per page.
- Columns: quote number with revision, date, company chip (internal only, never in the PDF), customer, net amount, valid until, status chip (icon, text and colour).
- Row actions: View, Edit, Revise, Duplicate, Print or Download PDF, Mark as Sent, Mark Negotiation, Mark Lost, Mark Cancelled, Convert to PO, Delete. Show only the actions allowed by 6.7 for that status (disable with a tooltip explaining why).
- **Duplicate** creates a new Draft with a new number from any quotation, optionally for a different company or customer.
- **View page** shows the read-only A4 preview, the status timeline (history and audit), all actions, Download PDF, Print, and Share: a WhatsApp link (`wa.me` with a short message), an email link (`mailto:` with subject and body), and the device share sheet where the browser supports sharing files. Tell me clearly that email and WhatsApp links cannot attach the PDF by themselves.

7.5 Other pages

- **Customers:** list, create, edit, soft-delete, restore, Excel import. A detail drawer shows quotation history, totals, and default T&C per company.
- **Items and Rates:** item list with search and filters (material, active); item drawer with details; a Rates tab with this company's rate per unit, inline edit, and a rate history table; Excel import; active or inactive toggle.
- **T&C Presets:** manage headings and preset wording (6.11).
- **Reports** (each with company and date filters and Excel export): Quotation Register, Customer-wise Summary, Item-wise Summary (quantity and value per item), Pending Follow-up, and a GST Summary of converted quotations (taxable, CGST, SGST, IGST; labelled 'for reference, confirm with your CA').
- **Recycle Bin:** as in 6.9.
- **Settings** (tabs): Companies (a tab per company: Profile, Bank, Branding files with preview, Numbering and defaults, PDF theme preview), Dropdown Manager, Security (inactivity timeout, change passcode), Backup and Export (Excel of all data per company), Activity Log (read-only audit trail with filters), About (version). Show a 'profile complete' checklist per company.

8. Quotation form: every field and behaviour

8.1 Layout

- Two panes: the form on the left in section cards, and a **live A4 preview** on the right that uses the same PDF component as section 9 (scaled to fit, with zoom and a Colour or Light/B&W switch). Below 1280 px the preview moves to a 'Form | Preview' tab switch.
- A sticky bottom bar always shows the Net Amount, the status chip and the auto-save indicator, with buttons: Save Draft, Preview PDF, Mark as Sent.
- Section cards, in order: 1 Quote Details, 2 Customer, 3 Items, 4 Charges and Totals, 5 Terms and Conditions, 6 Remarks. Collapsible cards; an error badge on a card header when it has errors.

8.2 Card 1: Quote Details

Field	Behaviour
Quote From	Large selector showing both companies with logo and colour. Changing it re-themes the form and the preview, and reloads that company's bank, default T&C, prefix and validity. Locked after the first save
Quote No	Read-only, for example `KPI/2026-27/0001`. Shows 'Will be assigned when you save' before the first save
Revision	Read-only 'R0', 'R1' and so on. New revisions are created only through Revise
Quote Date	Date picker in DD/MM/YYYY, default today
Validity	Dropdown: 7 Days, 15 Days, 30 Days, Custom Date. Shows 'Valid until DD/MM/YYYY'
Status	Read-only chip

8.3 Card 2: Customer

Field	Behaviour
Customer Name	Autocomplete (6.12); required
Contact Person or Kind Attention	Optional text
Phone	Validated (6.13)
Email	Validated
Requirement Via	Searchable dropdown with '+ Add new option': Mail, WhatsApp, Person, Phone Call, IndiaMART, Reference, Other. It records how the enquiry reached me
Address	Multi-line
City	Text

Field	Behaviour
State	Searchable dropdown of all states and union territories
PIN Code	6 digits
GSTIN	Upper-cased, validated, with a green tick or an inline message
Place of Supply	Defaults from GSTIN or State; editable. A badge shows 'Intra-state: CGST + SGST' or 'Inter-state: IGST'
Reference	Text: the enquiry reference or the person who referred the customer
Subject	Optional line printed under the title, for example 'Quotation for LDPE bags'
Delivery address	Optional collapsible block; printed only when filled

8.4 Card 3: Items

Render each item as an **expandable card** with a summary line (number, code, name, merged size, quantity and unit, rate, amount), not as a very wide table. Inside each card the fields are laid out in a tidy grid:

1. Item Code (picker) and Item Name (picker). Selecting from the master fills the rest.
2. Specification (text area, 2 rows).
3. Material (dropdown).
4. Size 1: value and unit. Size 2: value and unit. A '+ Size 3' toggle reveals the optional third size. A **read-only Merged Size box** shows the result of 6.3.
5. GSM, Micron or Gauge: value and spec unit.
6. Print: Plain or Printed, and the number of colours when Printed.
7. HSN code (text, validated as 4 to 8 digits).
8. Quantity: value and unit (Kg, Pcs, Nos, Box, Rolls, Meter, Sq. Meter).
9. Rate: number, with the unit label shown automatically ('₹ / Kg'), and the hint line from 6.12 ('Master (KPI): ... Last to this customer: ...').
10. GST %: dropdown, defaults from the item.

11. Line Amount: read-only, live.

Behaviour:

- Buttons: '+ Add New Item Row', Duplicate row, Delete row (with a 5-second Undo toast instead of a confirm), Move up, Move down. Maximum 100 rows.
- Keyboard: Tab moves to the next field in the order above. Pressing Enter in the last field of a row adds a new row and puts the cursor in its Item Code. Pressing Escape closes an open dropdown.
- Row add and delete animate with a smooth expand and collapse (200 ms).
- A new quotation starts with one empty row.

8.5 Card 4: Charges and Totals

- Discount: choose % or ₹, then enter the value.
- Freight or packing: editable label, ₹ amount, GST rate.
- Round-off: on or off (default from the company setting).
- **Totals panel** (live): Subtotal, Discount, Freight, Taxable Total, then CGST and SGST (or IGST) listed per GST rate, Round-off, **Net Amount** in a gradient highlight, and **Amount in Words** (automatic, read-only).

8.6 Card 5: Terms and Conditions

Implement exactly 6.11: 8 default rows, a heading dropdown, a preset dropdown, an editable body, On/Off checkbox, reorder, Add row up to 12, and 'Save as this customer's default for this company'.

8.7 Card 6: Remarks

- **Remarks** (printed under the totals, optional).
- **Internal note** (never printed; shown only inside the app with a 'Private' label).

8.8 Saving, validation and safety

- Behaviour for saving, autosave, unsaved-changes warning, master update prompts and conflicts: 6.12.
- Inline validation as in 6.13, shown as soon as a field loses focus; the Mark as Sent button shows a summary of everything still missing.

- Every number input uses the same parsing and formatting helpers as the domain library, so the screen, the PDF and the database always show the same figures.
- The form works with the keyboard only, on a laptop, tablet and phone.

9. The two A4 quotation PDFs

Build two completely different, rich A4 designs. Neither may look plain. Both take the same data object (`QuotationPdfData`) plus a mode (Colour or Light/B&W), and are chosen through a template registry keyed by the company's `pdf_theme` , so a third template can be added later without touching other code.

9.1 Rules common to both designs

- Page 210 mm by 297 mm. Colour bands and ornaments may reach the page edge, but all text and data stay at least 10 mm inside the edge, because printers cannot print the last few millimetres.
- Page 1 has the full header. Later pages have a compact header (company name, quote number, 'Page X of Y'). Every page has a slim footer with the company's own contact line and 'Page X of Y'.
- Content order: header, title, two information cards, optional subject line, items table, totals with amount in words, remarks, Terms and Conditions, bank details, signature and seal block.
- **Information cards.** Left card 'Quotation To': customer name, kind attention, address, city, state, PIN, GSTIN, phone, email (delivery address when filled). Right card: Quote No, Date, Revision, Valid Until, Reference, Place of Supply with the tax type. 'Requirement Via' and the internal note are never printed.
- **Items table columns:** S.No, Item Code, Description (name, specification, material, print), Size (merged), GSM, Micron or Gauge, HSN, Quantity with unit, Rate with unit (for example ₹/Kg), GST % (only when the quotation has more than one GST rate), Amount. The amount column header carries the ₹ sign. Numbers are right-aligned with tabular numerals.
- **Pagination.** The table header repeats on every page. A row never splits across pages. The totals block stays together. The signature block can never sit alone on a page: if needed, move the last Terms rows with it.
- **Type.** Body 9.5 pt, table 9 pt, Terms 8.5 pt, nothing below 8 pt. Use Noto Sans (or Inter) for body text, numbers and amounts, and confirm the ₹ glyph renders in every font that shows an amount. Embed all fonts.

- **Contrast.** Any text on a gradient must meet WCAG AA contrast (4.5:1 for normal text, 3:1 for large bold text). Put white text only on the darker stops; put dark ink on the lighter stops.
- **Modes.** Colour (default) and Light/B&W (no gradients, textures or illustrations; black, grey and thin rules). Set `print-color-adjust: exact` and `-webkit-print-color-adjust: exact` so backgrounds print without the user ticking 'Background graphics'.
- **Watermarks:** Draft shows DRAFT, Superseded shows SUPERSEDED, Cancelled shows CANCELLED, Expired shows EXPIRED; diagonal, centred, opacity 8% or less. Other statuses have none.
- **File hygiene.** Target under 1 MB for a typical quotation (hard ceiling 2 MB). Resize logos to at most 600 px wide before embedding. Do not use SVG filters, large blurs or photographs in print templates, because they become heavy raster images. Build textures from vector paths or tiny repeating tiles at 4 to 6% opacity.
- **Metadata and isolation.** PDF Title: 'Quotation [number] - [customer]'. Author: the issuing company's display name. No string, colour, image name or font name from the other company may appear anywhere in the file (test in 13.9).
- **Missing images.** If there is no logo, draw a monogram of the company's initials in that company's own theme. If there is no seal or signature, show a labelled empty frame. Never show another company's artwork.
- **All illustrations are original SVG** drawn by you. No stock images, no copied artwork.

9.2 Design A: Khandelwal Plastic Industries (Ocean Aurora)

Mood: fresh, industrial, eco-aware. Palette: navy #0B2A5B, blue #1E88E5, teal #14B8A6, mint-green #4ADE80, ink #0B2540, sky tint #EAF4FF, mint tint #E8F7EE.

Part	Design
Header	Full-width gradient band about 42 mm tall (navy to blue to teal to green) with a flowing wave along its lower edge. A glossy plastic-film sheen (diagonal highlight, 8 to 12%), fine wave lines and small bubbles. At the right, low-opacity original illustrations: film roll, pouch, hexagon molecule cluster and a small leaf. The logo sits on a white rounded plate at the left. Company name large in white (Outfit), then address, GSTIN, phone and email in white small text over the darkest part of the gradient
Title	'TAX QUOTATION' in a green gradient ribbon that overlaps the header's lower edge
Information cards	Two rounded cards: sky-to-mint soft gradient, a green left border, small round icons before each label
Items table	Header row gradient from navy to deep blue to deep teal with white bold text. Body rows alternate white and a very light mint-blue. Thin light dividers
Totals	Right-hand card. Net Amount in a bold green-teal gradient band with large white text. Amount in words in a mint strip with dark ink text
Page watermark	Very faint hexagon grid (4 to 5%) in the centre
Lower part	Bank details card first (gradient border), then Terms and Conditions with numbered round green badges
Footer	Inverted wave band in the brand gradient with the contact line
Seal and signature	Bottom right inside a rounded frame: 'For Khandelwal Plastic Industries', seal and signature images, 'Authorised Signatory'
Fonts	Outfit for headings, Noto Sans for text and numbers

9.3 Design B: Satyam Polymers (Rose Silk)

Mood: warm, elegant, premium. Palette: coral #EF7D84, blush #F9B8BC, peach #FFD9BF, cream #FFF6E8, ivory #FFFDF8, rose tint #FBDADC, sand #F6E7D3, maroon ink #7A2E36. Never use white text on light coral.

Part	Design
Header	Cream and ivory panel with a silk-like gradient (coral to blush to peach to cream) sweeping in from the top corners as arches and flowing curves, with a rose-gold shimmer. The logo sits inside a light-red ring. Company name in an elegant serif (Playfair Display) in deep maroon, then address, GSTIN, phone and email in maroon small text. Small original ornaments (granule dots, molecular chains) in the corners. A paper or linen grain tile at 5% or less
Title	'QUOTATION' centred, with a thin double rose-gold line beneath and a small diamond in the middle
Information cards	A row of rounded cream chips with light-red borders for Quote No, Date, Revision and Valid Until, then two cards with an ivory-to-sand fade, light-red outline and small corner ornaments
Items table	Header row in a blush-to-coral tint gradient with **maroon** bold text. Body rows alternate cream and white. Thin light-red horizontal rules only, no vertical lines
Totals	Cream card. Net Amount in a rose gradient pill with deep maroon text. Amount in words in an italic serif line
Page watermark	Very faint arch pattern
Lower part	Terms and Conditions first, with small diamond bullets, then the Bank details card with a light-red gradient outline

Part	Design
Footer	Scalloped (garland-like) edge fading from cream to rose, with the contact line
Seal and signature	Seal in a round dashed frame on the left; signature on the right with 'For Satyam Polymers' and 'Authorised Signatory'
Fonts	Playfair Display for headings, Noto Sans for text and numbers

9.4 Actions and verification

- Buttons: Preview, Download PDF, Print. The PDF is always generated again from the stored quotation data, so it is repeatable.
- Verify in Phase 5 with fixtures: 1 item, 8 items, 25 items (multi-page), long specifications, mixed GST rates, discount and freight, Draft and Superseded watermarks, Light/B&W mode. Check page count, no clipped text, repeated table header, correct 'Page X of Y', ₹ rendering, file size, contrast and the other-company text scan. Give me page images for approval.
- I will test a real colour print and a black-and-white print myself; list this under 'I must test'.

10. Design system: themes, gradients, textures, fonts and animation

The look is a modern enterprise ERP with a rich, polished feel. I want **gradients and a subtle texture on every surface** except the places listed in 10.4. Nothing in the app may look plain or default.

10.1 Shared tokens

Token	Light mode	Dark mode
Page background	#F8FAFC with a soft brand-tinted mesh at the top	#0B1220
Card	#FFFFFF, 12 px radius, soft two-layer shadow	#111A2E
Border	#E2E8F0	#1F2A44

Token	Light mode	Dark mode
Text	#0F172A and #475569	#E2E8F0 and #94A3B8
Sidebar	#0F172A with a deep navy gradient; the active company colour glows faintly from the bottom	same, slightly deeper

Default to the system light or dark preference; a toggle overrides it and is remembered. In dark mode, deepen the gradients and halve the texture opacity.

10.2 Company themes (set through CSS variables on `data-company`)

Define `--brand-1` to `--brand-4` (gradient stops), `--brand-ink`, `--brand-tint-1`, `--brand-tint-2`, `--brand-gradient`, `--brand-gradient-soft`, `--brand-glow`, `--brand-action-1`, `--brand-action-2` (button gradient) and `--texture`.

	Khandelwal: Ocean Aurora	Satyam: Rose Silk
Gradient stops	#0B2A5B, #1E88E5, #14B8A6, #4ADE80	#EF7D84, #F9B8BC, #FFD9BF, #FFF6E8
Ink (text)	#0B2540	#7A2E36
Tints	Sky #EAF4FF, Mint #E8F7EE	Rose #FBDADC, Sand #F6E7D3, Ivory #FFFDF8
Button gradient	Deep blue to deep teal (white text)	Deep rose, about #B83F4B to #C94A55 (white text; verify 4.5:1 and darken if needed)
Glow	Blue-teal	Rose-gold
Textures	Plastic-film sheen, wave lines, bubbles, hexagon grid	Paper and linen grain, granule dots, silk curves
Mesh (login, banners)	Blue, teal and mint soft blobs	Rose, peach and cream soft blobs

All Companies mode uses a neutral indigo accent (#4F46E5). **Never put white text on light coral.** Small text on Satyam surfaces is maroon.

10.3 Where gradients and textures appear

Login page (full mesh gradient with a glass card and faint grain); sidebar; topbar (glass blur with a thin gradient line); Home banner; Dashboard header; KPI cards (soft tinted gradient, gradient icon chip, sparkline); primary buttons (with a hover shine and a press effect); chart bars and areas (fade from top to bottom); card hover (gradient border); skeleton loaders (moving gradient shimmer); toasts, toggles, progress bars and avatars; active menu indicator; empty-state illustrations; modal headers. Build textures from inline SVG or CSS only, keep them light, and pause floating animations when off-screen.

10.4 Where gradients and textures must NOT appear

Table data rows, form inputs, text areas, dropdown lists, any text smaller than 14 px, status chips, and any place where reading an exact number matters. These stay flat with high contrast.

10.5 Status chips (solid colour, always icon plus text, never colour alone)

Status	Colour	Icon
Draft	Slate #64748B	pencil
Sent	Blue #2563EB	send
Negotiation	Amber #D97706	message circle
Converted	Emerald #059669	check circle
Lost	Crimson #B91C1C	x circle
Cancelled	Grey #6B7280	ban
Expired	Brown #9A3412	clock
Superseded	Violet #7C3AED	layers

Crimson for Lost must stay clearly different from Satyam's light coral, and Emerald for Converted clearly different from Khandelwal's green; the icon and text make each chip unmistakable.

10.6 Typography

Plus Jakarta Sans for the interface (14 px base, nothing below 12 px). Outfit for headings (600 and 700). Tabular numerals (`font-variant-numeric: tabular-nums`) for every amount, quantity and quote number. Self-host fonts with `next/font` (no runtime font CDN). PDF fonts are defined in section 9.

10.7 Components

- Buttons: primary (brand gradient), secondary (outline), ghost, danger (solid crimson). Height 40 px, 44 px touch target on mobile.
- Inputs: solid white, 1 px border, 2 px brand focus ring, inline error text under the field.
- Searchable dropdowns: type to filter, arrow-key navigation, the last row is '+ Add new option'.
- Tables: sticky header, hover tint, numbers right-aligned, flat rows.
- Destructive actions: a confirmation dialog with the item name; permanent delete and reversing a conversion need typed confirmation.
- Toasts: bottom right, auto-close after 4 seconds, with an Undo button where it applies.

10.8 Animation (150 to 300 ms unless stated)

Where	Animation
Page change	Fade plus 8 px slide, 200 ms
Login card	Fade and slide up, 300 ms
KPI numbers	Count up from 0, 600 ms
Charts	Draw-in, 600 to 800 ms
Card hover	Lift 2 px with a brand glow, 150 ms
Loading	Skeleton with moving gradient shimmer, 1.2 s loop
Item row add or delete	Height expand or collapse, 200 ms
Toast	Slide in, 200 ms
Dropdowns, menus	Fade and scale, 150 ms
Dialogs	Fade and scale, 200 ms

Where	Animation
Status chip change	Colour cross-fade, 250 ms
Button press	Scale to 0.97, 100 ms
Sidebar collapse	Width change, 250 ms
Company switch	Theme colour transition, 300 ms
Illustrations	Slow floating loops, 6 to 10 s, transform only

Animate only transform, opacity and colour. Respect `prefers-reduced-motion` (replace motion with instant changes). Pause loops when the tab is hidden. **No animation in print or PDF.**

10.9 Accessibility and responsive behaviour

- WCAG AA contrast, visible focus rings, labels on every field, `aria-live` for toasts and validation messages, a skip-to-content link, and a 'Keyboard shortcuts' list under Settings (Ctrl+K search, Ctrl+B sidebar, Ctrl+S save draft).
- Breakpoints: 1280 px and above (two-pane form), 1024 to 1279 px (preview in a tab), 768 to 1023 px (drawer sidebar), below 768 px (single column; tables turn into cards).
- Dark mode must cover every screen and chart. The PDFs are always light.

10.10 Illustrations

Create an original SVG set that uses the theme gradients through CSS variables: login brand panel (film roll, pouch, molecule), the two Home banners, empty states (no quotations, no customers, no items, empty Recycle Bin, no search results), loading, error and 404 pages. No stock images and no copied artwork.

11. Delivery plan, Phases 0 to 5

Follow rule 1.2: one phase at a time, a Phase Report at the end of each, then wait for 'continue'. 'Done when' lists what **you (the AI)** must run and prove. 'I check' lists what **I** will test by hand.

Phase 0: Preparation (mostly my work)

Build. A checklist document `docs/PHASE0-CHECKLIST.md` in simple English with:

- Accounts I must create, each with 2-step verification: GitHub, Supabase, Vercel. Say which are free and which plan I should check before launch.
- For each company (mark each item 'needed now' or 'can be added later in Settings'): full name, address, GSTIN, PAN, phones, email, website; bank name, account name, account number, IFSC, branch; logo, seal and signature as PNG with transparent background; my 7 to 8 real Terms and Conditions; one old quotation (customer name removed) as a sample of my format.
- For both: my Login ID and recovery email; 20 to 30 main items with rates for each company; 10 main customers; confirmation of the prefixes KPI and SP.
- **Questions for my CA:** which GST rates and HSN codes apply to my products; how freight and packing is taxed; whether round-off is acceptable; whether e-invoicing will apply to me later.

Done when. The checklist exists and is clear. I may continue with placeholders for anything I do not have yet.

Phase 1: Design preview (look only, no database)

Build. A clickable preview with dummy data and the real design system (section 10): the login page; the app shell with a working collapsible sidebar and a company switcher that really re-themes the whole app between Khandelwal and Satyam and All Companies; the Home banner, KPI cards and an empty-state illustration; a Dashboard layout with dummy charts; **both A4 quotation designs** side by side, each with 5 dummy item rows and one with 25 rows; light and dark mode; the animations from 10.8.

Done when. Screenshots on desktop, tablet and phone are delivered; contrast results for the gradient text are listed; the ₹ sign renders in every font; reduced-motion works; no real data or database is used.

I check. Do the gradients and textures look rich and modern? Is Satyam readable? Does the sidebar collapse smoothly (and with Ctrl+B)? Does dark mode look right? I reply 'design approved' or list changes. **The design is locked after approval.**

Phase 2: Foundation (project, database, login)

Build. The project and folder structure (3.3); `.env.example`; SQL migrations for every table in 4.1; the functions in 4.2; Row Level Security; seed data (4.4); login, lockout, inactivity logout and route protection (section 5); the real app shell wired to the database;

company choice remembered across refreshes; CI that runs lint, type check and tests. Provide click-by-click steps for creating the Supabase **development** project (Mumbai region), running migrations, and setting environment variables.

Done when. Migrations run cleanly on an empty database; the RLS test proves anonymous access returns nothing; the 50-parallel-call numbering test passes; wrong-passcode lockout and inactivity logout work; sign-up is disabled; CI is green.

I check. Log in; fail 5 times and see the lock; log out; refresh and see that the company choice stays.

Phase 3: Calculation engine and tests

Build. Pure functions in `lib/domain` for everything in 6.1 to 6.5, the status rules in 6.7 and the validators in 6.13: money helpers, Indian number formatting, amount in words, size merger, unit rules, GST and totals engine, financial year and numbering, status transition rules. Write every test in section 13.

Done when. All section 13 tests pass, coverage of `lib/domain` is at least 95%, and the report shows the real test output. List every assumption (for example freight GST) under 'Please confirm with the CA'.

I check. Run `npm test` and read the results. My CA checks the worked example in 13.3.

Phase 4: Masters and Settings

Build. Settings with a tab per company (Profile, Bank, Branding files with preview, Numbering and defaults, PDF theme preview), Dropdown Manager, Security, Backup and Export, Activity Log; Customers; Items and Rates with per-company, per-unit rates and a rate history viewer; T&C Presets; Recycle Bin for masters; Excel import for customers and items with a downloadable template and a **dry-run preview that shows row-by-row errors before anything is saved**; Excel export. Skeletons, empty states, toasts, animations, audit entries and a 'profile complete' checklist per company.

Done when. CRUD works on every master; one item can have different rates for the two companies; uploads reject wrong or oversized files; a bad import file is caught in the dry run; the exported Excel opens correctly; the RLS and unit tests still pass.

I check. Enter my real data for both companies, upload logos, seals and signatures, add a few items with different rates per company.

Phase 5: The two A4 PDF templates

Build. Both designs exactly as in section 9, as React components fed by the same data object; the PDF pipeline decision (3.2) with a test against Vercel limits; Preview, Download and Print; all watermarks; Light/B&W mode; fixtures for 1, 8 and 25 items.

Done when. Every check in 9.4 passes automatically (page count, repeated header, Page X of Y, ₹ rendering, file size, no clipped text, contrast, other-company text scan); page images of every fixture are delivered; you explain which PDF method you chose and why.

I check. Approve the page images. Print in Chrome without ticking 'Background graphics' and confirm the gradients still print. Test a real colour print and a black-and-white print and confirm 25 items continue properly on page 2.

12. Delivery plan, Phases 6 to 10

Phase 6: Quotation form

Build. The New and Edit Quotation page exactly as in section 8 and the rules in section 6: all cards and fields, the item cards with Tab and Enter behaviour, the item and customer pickers, master save and update prompts, the 'last rate to this customer' hint, charges, live totals and amount in words, the 8-row T&C editor with presets and customer defaults, the live A4 preview using the Phase 5 templates, local crash recovery then autosave, the unsaved-changes warning, inline validation with the Mark as Sent summary, the sticky total bar, and saving with the atomic quote number and company lock.

Done when. The worked example in 13.3 entered through the form shows Net Amount 8,37,800.00 and the same figure in the preview and the saved record; an automated browser test creates one KPI and one SP quotation; the form works by keyboard alone; saving twice never creates two numbers; the browser test of the unsaved-changes warning passes.

I check. Recreate my old sample quotation and compare the totals and the PDF with my original. Test a Kg item, a Sq. Meter item and a Pcs item; same-state and other-state customers; a customer with saved T&C.

Phase 7: Quotations list and actions

Build. The list and view pages as in 7.4, and every action with the rules of 6.6 to 6.10: status tabs, search, filters, sorting, pagination, Edit (with the 'already sent' prompt), Revise, Duplicate, Delete and Restore, Convert to PO and Reverse conversion, Mark Sent,

Negotiation, Lost (with reason) and Cancelled, the daily expiry job plus the read-time fallback, the status timeline, Share links, and an audit entry for every change.

Done when. Automated tests prove: every illegal status transition is rejected by the **server**; Revise marks the old revision Superseded in one transaction and counts only the latest; Converted quotations cannot be edited or deleted; Delete and Restore return the right status; expiry works with and without the job; the delete dialog's two options behave as in 6.9.

I check. Revise a quotation and confirm R0 becomes Superseded and amounts are not counted twice. Convert one to PO and try to edit it. Delete one and restore it from the Recycle Bin.

Phase 8: Home, Dashboard and Reports

Build. Home (7.2), Dashboard (7.3) with the exact definitions of 6.14, the five Reports with Excel export (7.5), the command palette search, notification bell, and all animations (count-up, chart draw-in, skeletons).

Done when. The dashboard test fixture in 13.7 gives exactly the expected figures; a generated database of 10,000 quotations keeps the quotation list under 500 ms and the Dashboard under 1 second (add indexes or summary queries if needed and report the timings); every chart and card has loading, empty and error states; the company filter changes every number and colour.

I check. Make 10 to 15 real quotations in different statuses, work out the totals by hand, and compare with the Dashboard. Switch company and date filters.

Phase 9: Polish, security and quality assurance

Build and review. Go through every screen and fix what you find: animation, loading, empty and error states, phone and tablet layouts, dark mode, reduced motion, keyboard use, contrast. Run an accessibility scan (axe) and Lighthouse. Security review: RLS on every table, sign-up off, no secret in browser code, the service-role key used only where listed in 4.3, signed URLs that expire, security headers, rate limiting, XSS test strings in every free-text field (customer name, T&C, remarks, subject) rendering as plain text in the app and the PDF, SQL injection strings in search boxes, lockout, dependency audit with no high or critical issues. Performance: debounce, pagination, bundle size. Run every test suite, including the Playwright end-to-end suite in 13.10.

Done when. You deliver a table 'Problem found, severity, fix, status'. No critical or high problem is open. All tests pass. Remaining risks are listed honestly.

I check. Use the whole app on my phone and on a laptop; try dark mode; switch both companies; print both PDFs.

Phase 10: Deployment and launch

Build. Everything in section 14: the full deployment guide `docs/DEPLOYMENT.md`, the production environment variable list, the production seed script, the smoke-test checklist, backup and restore instructions, `README.md`, `ARCHITECTURE.md`, `RUNBOOK.md` and `CHANGELOG.md`. Do every step you can do with your tools, and for the rest give me click-by-click instructions and wait for my confirmation after each numbered step.

Done when. The app is live on a Vercel address (and my custom domain if I have one); the production smoke test in 14.4 passes; the launch gate in 14.6 is complete; you hand over the final report with the links, the list of what is configured where, and the post-launch routine from 14.7.

I check. Run the smoke test myself on the live site, then follow the launch gate.

13. Mandatory tests with expected results

Write these tests exactly (more are welcome). A phase is not finished until its tests pass. Never change an expected value to make a test pass without telling me why. All amounts are in ₹ unless stated.

13.1 Amount in words

Amount	Expected text
0	Rupees Zero Only
1	Rupees One Only
100	Rupees One Hundred Only
44,000	Rupees Forty Four Thousand Only
1,00,000	Rupees One Lakh Only
1,25,000	Rupees One Lakh Twenty Five Thousand Only
8,37,800	Rupees Eight Lakh Thirty Seven Thousand Eight Hundred Only

Amount	Expected text
1,00,00,000	Rupees One Crore Only
12,34,567.50	Rupees Twelve Lakh Thirty Four Thousand Five Hundred Sixty Seven and Fifty Paise Only
1.50	Rupees One and Fifty Paise Only
0.50	Fifty Paise Only
2,50,00,00,000	Rupees Two Hundred Fifty Crore Only
99,99,99,999.99	Rupees Ninety Nine Crore Ninety Nine Lakh Ninety Nine Thousand Nine Hundred Ninety Nine and Ninety Nine Paise Only

13.2 Money, line amounts and Indian formatting

Input	Expected
Format 12345650 paise	1,23,456.50
Format 0 paise	0.00
Format 99 paise	0.99
Format 100000000 paise	10,00,000.00
Format 123456789 paise	12,34,567.89
Format -50050 paise	-500.50
Display rate 1,420,000 (rate_e4)	142.00
Display rate 3,750 (rate_e4)	0.375
Display rate 1,234,500 (rate_e4)	123.45
5000 Kg × ₹142	7,10,000.00
100000 Pcs × ₹0.375	37,500.00
2.5 Kg × ₹33.33 (exactly 83.325)	83.33 (half up)
0 quantity × any rate	0.00

Input	Expected
Rounding half up: 0.5, 1.5, 2.5 paise-fractions	1, 2, 3

13.3 GST and totals (the worked example is case A)

Company state code 23 is used here only as a test value.

Case	Input	Expected
A. Worked example, same state	5000 Kg × ₹142, GST 18%, customer in the company's state	Subtotal 7,10,000.00; CGST 9% 63,900.00; SGST 9% 63,900.00; round-off 0.00; **Net 8,37,800.00**; words 'Rupees Eight Lakh Thirty Seven Thousand Eight Hundred Only'
B. Same, other state	Same lines, customer in another state	IGST 18% 1,27,800.00; Net 8,37,800.00
C. Mixed rates, same state	Line 1 ₹1,00,000 at 5%; line 2 ₹2,00,000 at 18%	5%: CGST 2,500.00 and SGST 2,500.00; 18%: CGST 18,000.00 and SGST 18,000.00; total GST 41,000.00; Net 3,41,000.00
D. Discount 10% on case C	Discount 30,000.00	Allocated 10,000.00 and 20,000.00; taxable 90,000.00 at 5% and 1,80,000.00 at 18%; GST 4,500.00 and 32,400.00; total GST 36,900.00; Net 3,06,900.00
E. Freight on case A	Freight ₹1,000.00 at 18%	Taxable 7,11,000.00; CGST 63,990.00; SGST 63,990.00; Net 8,38,980.00
F. Round-off up	1 Pcs × ₹1,000.50 at 18%, same state	CGST 90.05; SGST 90.05; gross 1,180.60; round-off (+) 0.40; Net 1,181.00
G. Round-off down	1 Pcs × ₹1,000.34 at 18%, same state	CGST 90.03; SGST 90.03; gross 1,180.40; round-off (−) 0.40; Net 1,180.00

Case	Input	Expected
H. Exact half	1 Pcs × ₹100.50 at 0%	Gross 100.50; round-off (+) 0.50; Net 101.00
I. Discount tie	Three lines of ₹100.00 at 0%, 5% and 18%; discount ₹10.00	Allocations 3.34 (0%), 3.33 (5%), 3.33 (18%), summing to exactly 10.00
J. Rejected discount	Discount above the subtotal	Validation error
K. Full discount	Discount 100%	Taxable 0.00; GST 0.00; Net 0.00

Place of Supply tests. Company state 23: customer GSTIN starting 23 gives INTRA; starting 27 gives INTER; no GSTIN with state Madhya Pradesh gives INTRA; GSTIN starting 27 with Place of Supply changed to 23 gives INTRA; no Place of Supply blocks Mark Sent but allows a Draft. A company and Place of Supply both in a union territory without legislature (for example Chandigarh) shows UTGST instead of SGST.

13.4 Merged size

Size 1	Size 2	Size 3	Result
44 mm	44 mm	empty	44 mm x 44 mm
44 mm	empty	empty	44 mm
44.50 mm	empty	empty	44.5 mm
350 mm	450 mm	50 mm	350 mm x 450 mm x 50 mm
12 inch	18 cm	empty	12 inch x 18 cm
empty	44 mm	empty	Validation error
0 mm	empty	empty	Validation error
44 with no unit	empty	empty	Validation error

13.5 Financial year, numbers and file names

Input	Expected
Quote date 01/04/2026	FY 2026-27
Quote date 31/12/2026	FY 2026-27
Quote date 01/01/2027	FY 2026-27
Quote date 31/03/2027	FY 2026-27
Quote date 01/04/2027	FY 2027-28
KPI, 2026-27, number 1	KPI/2026-27/0001
SP, 2026-27, number 125	SP/2026-27/0125
KPI, 2026-27, number 10000	KPI/2026-27/10000
KPI/2026-27/0001, revision 1	KPI/2026-27/0001-R1
Revision 0	No suffix
File name for KPI/2026-27/0001-R1 and customer 'Shri Ram & Sons (P) Ltd.'	KPI-2026-27-0001-R1_Shri_Ram_Sons_P_Ltd.pdf

Numbering database tests: 50 parallel calls for one company and one financial year return the numbers 1 to 50, each exactly once, with no gaps; KPI and SP series are independent; a new financial year restarts at 1; the number of a deleted draft is never reused.

13.6 Validators and status transitions

Check	Valid	Invalid
GSTIN	27AAAAA0000A1Z5; 27aaaaa0000a1z5 (valid after upper-casing)	27AAAAA0000A1Z (14 characters); 00AAAAA0000A1Z5 (state code 00); 27AAAAA0000A0Z5 (13th character 0)
PIN	452001	052001; 45200; 4520011
Phone	9876543210; +91 98765 43210 (becomes 9876543210)	5876543210; 98765

Check	Valid	Invalid
Email	a@b.co	a@b; a b@c.com
PAN	ABCDE1234F	ABCD1234FG
IFSC	HDFC0001234	HDFC1001234
Bank account	123456789	12345678

Allowed transitions (the server must accept): Draft to Sent; Sent to Negotiation; Sent to Converted (with PO details); Negotiation to Lost (with reason); Converted to Negotiation through the Reverse conversion action only (reason required). Rejected transitions: Draft to Converted; Draft to Lost; Draft to Expired; Converted to Lost; Lost to Sent; Expired to Sent; Cancelled to Draft; anything out of Superseded; setting Superseded by hand.

13.7 Dashboard fixture (one company, all dates)

Create these rows with customers X, Y and Z, plus a few rows for the other company to prove the company filter.

Quote	Customer	Status	Net amount	Notes
Q1 R0	X	Superseded	1,00,000.00	Old revision; must be ignored
Q1 R1	X	Converted	95,000.00	PO value 95,000.00; history Draft, Sent, Converted
Q2	X	Sent	2,00,000.00	valid_until = today + 2 days; reached Sent
Q3	Y	Negotiation	3,00,000.00	valid_until = today + 10 days; reached Sent
Q4	Y	Lost	50,000.00	reached Sent
Q5	Z	Cancelled	25,000.00	Cancelled from Draft; never Sent
Q6	Z	Draft	10,000.00	
Q7	Z	Expired	40,000.00	reached Sent
Q8	Y	Sent	5,00,000.00	In the Recycle Bin; must be ignored

Expected results:

Measure	Expected
Total quotations	7 quotations, ₹7,20,000.00 (includes 1 draft)
Converted to PO	1, PO value ₹95,000.00
Win rate	33.3% (1 ÷ (1 + 1 + 1))
Pending	2, ₹5,00,000.00
Lost or Cancelled	2, ₹75,000.00; Expired line: 1, ₹40,000.00
Expiring soon (3 days)	1, ₹2,00,000.00
Average quotation value	₹1,02,857.14
Funnel	Created 7, Sent 5, Converted 1
Top customers	Y ₹3,50,000.00; X ₹2,95,000.00; Z ₹75,000.00
Customer-wise win rate	X 100%; Y 0%; Z 0%

Also test: a date range that excludes some rows; the This FY boundary between 31/03 and 01/04; the company filter returning only that company's figures; All equalling the sum of both.

13.8 Database and security tests

- Migrations run on an empty database; the seed can run twice without duplicates.
- With the anonymous key and with no session, every table returns zero rows and every insert, update and delete fails. With the owner session, normal access works.
- A storage file without a signed URL cannot be read; a signed URL stops working after it expires.
- The sign-up endpoint is rejected.
- Lockout: after 5 failures the 6th attempt is refused even with the right passcode until 15 minutes pass (use a fake clock).
- XSS strings such as `<script>alert(1)</script>` and `"><img src=x onerror=alert(1)>` in every free-text field appear as plain text in the app and in the PDF. A string such as `' OR 1=1 --` in any search box returns normal results.

- Soft-deleted rows appear only in the Recycle Bin. A Converted quotation cannot be deleted through the API. Two tabs saving the same quotation produce a conflict warning through `row_version`.
- Every write produces an `audit_log` row.

13.9 PDF isolation and quality tests

For every fixture PDF of each company, extract all text and all metadata and assert:

- None of the **other** company's legal name, display name, GSTIN, PAN, bank account number, IFSC, phone, email, address fragments or quote prefix appear. The file name carries only its own prefix. Embedded font and image names contain no other company's name. PDF Title and Author name only the issuing company.
- The ₹ sign or the 'Amount (₹)' header is present. A typical fixture is under 1 MB (never over 2 MB). The 1-item fixture is 1 page; the 25-item fixture is at least 2 pages with the table header repeated and 'Page X of Y' correct.
- A Draft PDF contains the word DRAFT; a Sent PDF does not.
- The first-page images of Design A and Design B differ clearly (perceptual hash distance above a set threshold), which proves the two designs are really different.

13.10 End-to-end tests (Playwright)

1. Five wrong passcodes lock the account; the right passcode works after the lock ends; inactivity logs out.
2. The company switcher changes `data-company` and the accent colours, and the choice survives a refresh. The sidebar collapses by click and by Ctrl+B and the state survives a refresh; the quotation form collapses it automatically.
3. Create the worked example as a KPI quotation: Net 8,37,800.00 in the form, the preview and the saved record; number KPI/2026-27/0001 on a fresh database; the PDF downloads with the correct file name.
4. Create an SP quotation: its number is SP/2026-27/0001, independent of KPI.
5. Revise the KPI quotation: R1 is created, R0 becomes Superseded, and the Dashboard counts the quotation once.
6. Mark Sent, then Negotiation, then Convert to PO with PO details; the quotation is then locked.
7. Delete a quotation, find it in the Recycle Bin, restore it.

8. Type a new item in a quotation: the 'Save to master?' prompt appears and the item shows in Items and Rates. Change a rate: the 'Update master rate?' prompt appears and the rate history shows the change.
9. Customer autocomplete fills the fields; a customer's saved T&C appear in their next quotation for that company only.
10. Dashboard filters (company and date) change the numbers.
11. At a phone width: the sidebar is a drawer and the quotation form has a Form and Preview switch.
12. With reduced motion emulated, no animation runs and numbers appear instantly.
13. Printing hides the sidebar and topbar.

13.11 Test rules

Commands: `npm test` (unit and integration) and `npm run test:e2e`. CI runs everything on every push and blocks on failure. Coverage of `lib/domain` is at least 95%. Show me real output in every Phase Report.